UNIT IV – ARTIFICIAL INTELLIGENCE

WHAT IS A LANGUAGE MODEL?

- A language model (LM) is a statistical or neural network model that learns patterns in language and estimates the probability of a sequence of words or tokens.
- For example, given the sentence:

  "The sun rises in the ____"

- A language model may predict the word "east" because it is a likely continuation of the sentence.

Mathematically, a language model estimates:

$$P(w_1, w_2, \ldots, w_n)$$

where $w_1, w_2, \ldots, w_n$ represent words or tokens in a sequence.

Using the chain rule of probability:

$$P(w_1, \ldots, w_n) = \prod_{i=1}^{n} P\left(w_i \mid w_1, \ldots, w_{i-1}\right)$$

This means that the probability of a complete sequence can be calculated using the conditional probability of each word given the preceding words.

TYPES OF LANGUAGE MODELS

1. Statistical Language Models

These models estimate word probabilities from text data using statistical techniques.

- Unigram model: considers individual words.
- Bigram model: considers pairs of consecutive words.
- Trigram model: considers sequences of three words.
- Example: A bigram model predicts the next word using only the previous word.

2. Neural Language Models

These use neural networks to learn complex relationships between words.

Common architectures include:

- Recurrent Neural Networks (RNNs).
- Long Short-Term Memory networks (LSTMs).
- Gated Recurrent Units (GRUs).

They can learn contextual patterns that are difficult for simple statistical models to capture.

3. Transformer-Based Language Models

Transformers use attention mechanisms to learn relationships between tokens.

Examples include:

- BERT: designed primarily for contextual understanding.
- GPT: designed primarily for autoregressive text generation.
- T5: uses a text-to-text framework.
- MarianMT: a family of neural machine translation models.

IMPORTANT CONCEPTS IN LANGUAGE MODELS

- Tokenization: Dividing text into smaller units called tokens. A token may be a word, subword, character, or punctuation mark.

Example:

"I love NLP!" → ["I", "love", "NLP", "!"]

- Vocabulary: The set of tokens that the model recognizes.
- Token IDs: Numerical representations assigned to tokens so that a neural network can process them.
- Embeddings: Dense numerical vectors representing tokens. Similar words or related contextual uses may have similar representations.
- Context: The surrounding tokens that help determine the meaning of a particular token.
- Training: The process of adjusting model parameters using data to improve predictions.
- Inference: Using a trained model to make predictions or generate output for new input.

WHAT IS A LARGE LANGUAGE MODEL?

- A Large Language Model (LLM) is an artificial intelligence model trained on a very large amount of text data to understand language patterns and perform tasks such as answering questions, generating text, summarizing documents, translating languages, and writing computer programs.
- LLMs are generally based on deep learning, particularly the Transformer architecture. They learn statistical relationships between tokens during training and use these relationships to predict or generate text.
- Examples of popular LLM families include GPT, Llama, Gemini, Claude

- Input prompt:

  "What is artificial intelligence?"

- Possible output:

  "Artificial intelligence is a branch of computer science that develops systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, and understanding language."

- The LLM generates the response based on patterns learned during training and the context provided in the prompt.

FUNDAMENTAL CONCEPTS OF LLMS

- Natural Language Processing (NLP)
- Tokenization
- Token IDs - After tokenization, each token is mapped to a numerical identifier.
- Embeddings - An embedding converts a token ID into a dense numerical vector.
- Context and contextual embeddings
- Vocabulary and context window

Context and contextual embeddings

- The meaning of a word can depend on its surrounding words.

Consider:

"She deposited money in the bank."

"They sat on the river bank."

- The word "bank" has different meanings in the two sentences.
- Contextual language models use surrounding tokens to construct representations that help distinguish these meanings.

Vocabulary and context window

- Vocabulary: The set of tokens that a tokenizer can represent.
- Context window: The maximum amount of token context a model can process in a given operation, subject to the model's design and configuration.
- For example, a model with a context window of 8,192 tokens can process up to that many tokens in a supported input context. The available space may need to be shared between the prompt and generated output.
- A larger context window allows a model to consider more text, but it does not guarantee perfect understanding of every detail in a long document.

TRANSFORMER ARCHITECTURE IN LLMS

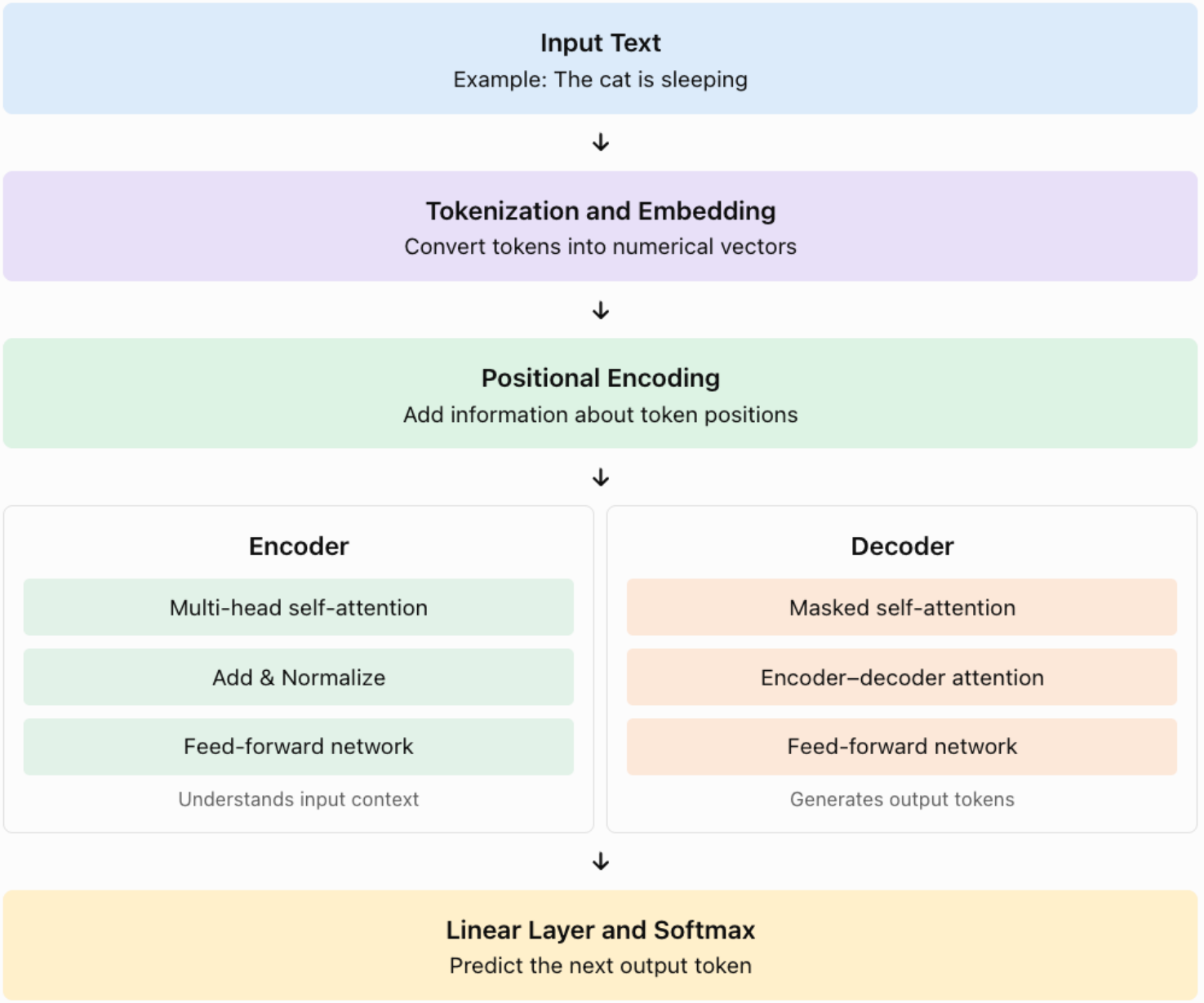
What is a Transformer?

- A Transformer is a deep learning architecture that uses attention mechanisms to identify relationships between tokens in a sequence.
- Most modern text-based LLMs use a Transformer-based architecture or a closely related design.

THE THREE MAJOR TYPES ARE:

Type	Description	Example
Encoder-only	Focuses on learning contextual representations	BERT
Decoder-only	Predicts the next token based on previous tokens	GPT
Encoder-decoder	Encodes an input and generates an output sequence	T5

TRANSFORMER ARCHITECTURE



MAIN COMPONENTS OF A TRANSFORMER

- Tokenization
- Embedding Layer
- Positional Encoding
- Self-Attention Mechanism
- Multi-Head Attention
- Add & Layer Normalization
- Feed-Forward Neural Network
- Encoder and Decoder
- Linear Layer and Softmax

1. TOKENIZATION AND EMBEDDING

Tokenization

- Divides input text into smaller units called tokens.
- Each token is assigned a numerical ID.

Example: I love AI → [I, love, AI]

Embedding Layer

- Converts token IDs into dense numerical vectors.
- Represents tokens in a form that neural networks can process.

Example: AI → [0.2, 0.5, -0.1, 0.7]

2. POSITIONAL ENCODING

- Provides information about the position of each token in a sequence.
- Helps the model distinguish different word orders.
- Positional information is combined with token embeddings.

Example:

- The dog chased the cat.
- The cat chased the dog.

Formula:

$$Z = X + P$$

X: Token embeddings; \(P\): Positional encodings.

3. SELF-ATTENTION MECHANISM

- Determines how much attention each token should pay to other tokens.
- Captures relationships between words, including distant words.
- Uses three vectors: Query (Q), Key (K), and Value (V).

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

Scaled Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

4. MULTI-HEAD ATTENTION

- Runs multiple attention operations in parallel.
- Each head can learn different relationships between tokens.
- Combines the outputs to produce richer contextual representations.

Process:

Input ⟶ Multiple Attention Heads ⟶ Concatenation ⟶ Linear Projection

$$\text{MultiHead}(Q, K, V) = \text{Concat}(head_1, \ldots, head_h)\, W_O$$

5. ADD & LAYER NORMALIZATION

- Residual connection: Adds the input to the sublayer output.
- Layer normalization: Normalizes activations to support stable training.
- Helps improve gradient flow in deep Transformer networks.

$$Y = \mathrm{LayerNorm}(X + F(X)))$$

Here, F(X) represents the output of an attention or feed-forward sublayer.

6. FEED-FORWARD NEURAL NETWORK

- Processes each token representation independently.
- Applies linear transformations and a nonlinear activation function.
- Learns complex patterns in the token representations.

$$FFN(x) = W_2\,\mathrm{ReLU}(W_1 x + b_1) + b_2$$

Purpose: Further transforms the information learned through attention.

7. ENCODER AND DECODER

Encoder

- Processes input tokens.
- Uses self-attention and feed-forward layers.
- Produces contextual representations.

Decoder

- Generates output tokens.
- Uses masked self-attention.
- In the original Transformer, uses encoder–decoder attention to access input representations.

Examples: Machine translation and text generation.

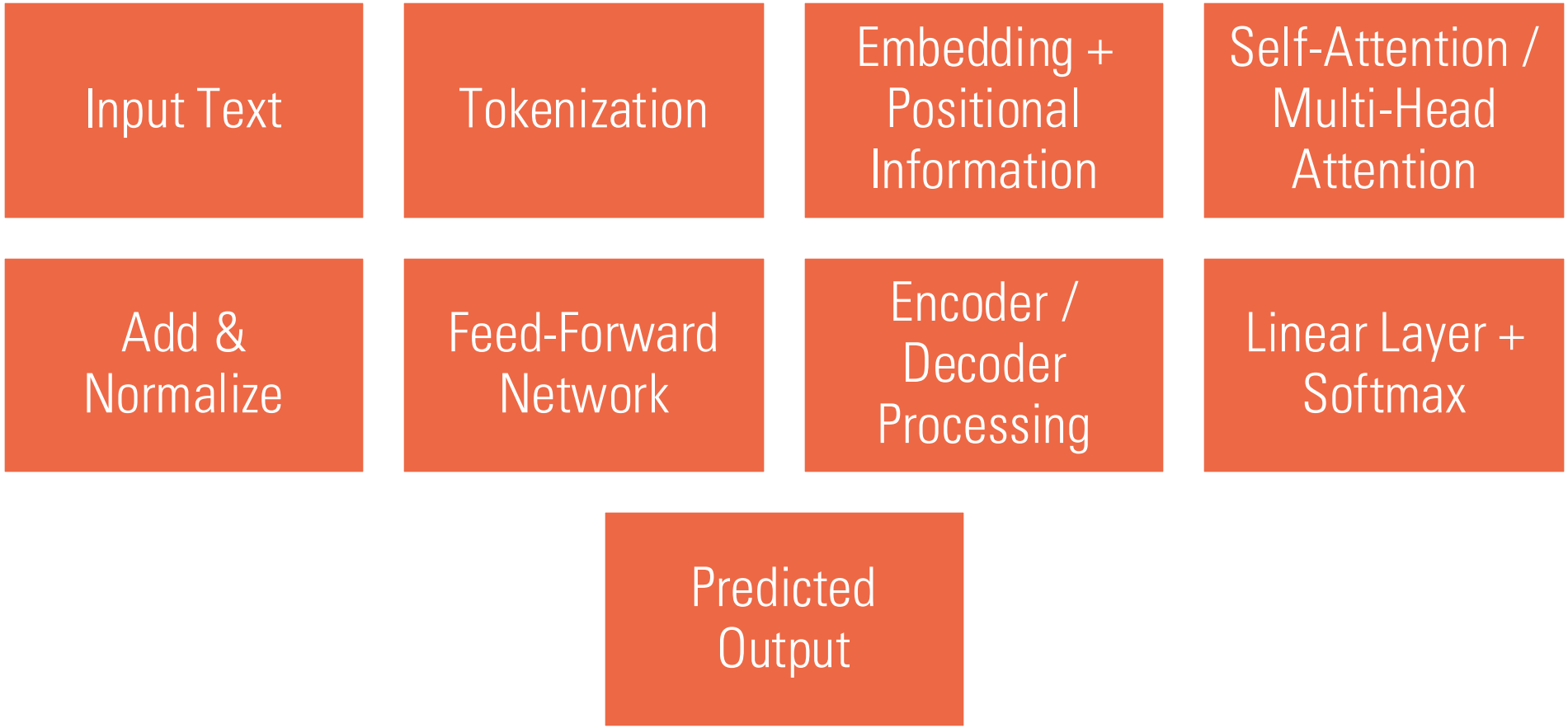
8. LINEAR LAYER AND SOFTMAX

- The linear layer converts the decoder representation into vocabulary scores (logits).
- Softmax converts these scores into probabilities.
- The model predicts the next token based on the resulting distribution.

$$P(w) = \frac{e^{z_w}}{\sum_j e^{z_j}}$$

Example: The model assigns probabilities to possible next words and selects or samples a token.

TRANSFORMER WORKFLOW



TEXT GENERATION

- Text generation is an NLP task in which a computer automatically produces meaningful human-readable text based on an input prompt, instruction, or context.
- Language models learn patterns from large amounts of text and use those patterns to predict the next token (word or part of a word).

HOW TEXT GENERATION WORKS?

1. Input Prompt

"Artificial intelligence is..."

↓

2. Tokenization and Embedding

Convert text into tokens and numerical representations.

↓

3. Transformer Model

Use self-attention and neural network layers to understand context and predict the next token.

↓

4. Token Prediction and Decoding

Select the next token and repeat the process.

↓

5. Generated Text

"Artificial intelligence is transforming many industries."

TYPES OF TEXT GENERATION

- Story and paragraph generation.
- Automatic essay and report writing.
- Question answering and chatbot responses.
- Code generation.
- Email and message drafting.
- Text summarization.

EXAMPLE

- Input prompt: Write a sentence about machine learning.
- Generated output: Machine learning enables computers to learn patterns from data and make predictions without being explicitly programmed for every task.

APPLICATIONS IN REAL LIFE

- ChatGPT and other conversational assistants.
- AI writing assistants.
- Automatic content creation.
- Programming assistants.
- Educational question and answer systems.

Common models: GPT, Llama and other decoder-only Transformer language models.

Limitation: Generated text may contain incorrect information, biased content, or fabricated facts. Therefore, important outputs should be verified.

SENTIMENT ANALYSIS

- Sentiment analysis is an NLP task that identifies the emotional tone or opinion expressed in a piece of text.
- It is commonly used to classify text into categories such as positive, negative, and neutral.

EXAMPLE



Review 1

"The product is excellent. I really love it!"

Positive



Review 2

"The product is disappointing and stopped working."

Negative



Review 3

"The product arrived yesterday."

Neutral

HOW SENTIMENT ANALYSIS WORKS USING TRANSFORMERS?

Input Text

"The movie was wonderful."

↓

Tokenization and Embedding

Convert the sentence into contextual token representations.

↓

Transformer Encoder

Self-attention captures the context and meaning of words.

↓

Classification Layer

Produces scores for positive, negative, or neutral sentiment.

↓

Predicted Sentiment: Positive

WHY ARE TRANSFORMERS USEFUL FOR SENTIMENT ANALYSIS?

- Traditional approaches often depend on predefined word lists or manually designed features. Transformers learn contextual representations, allowing them to interpret how words are used within a sentence.

For example:

  - "The food is good." — Positive.
  - "The food is not good." — Negative.

The word good appears in both sentences, but the word not changes the meaning. Contextual processing helps the model identify this difference.

TYPES OF SENTIMENT ANALYSIS

Type	Description	Example
Binary	Two sentiment classes	Positive / Negative
Ternary	Three sentiment classes	Positive / Negative / Neutral
Fine-grained	More detailed sentiment levels	Very negative to very positive
Aspect-based	Sentiment about a specific feature	Camera: Positive; Battery: Negative

Real-world applications

- Product review analysis in e-commerce.
- Customer feedback analysis.
- Social media monitoring.
- Brand reputation management.
- Movie and restaurant review classification.
- Customer support and satisfaction analysis.

Limitation: Sarcasm, irony, slang, mixed emotions, and domain-specific language can make sentiment classification difficult.

MACHINE TRANSLATION

- Machine translation is the automatic conversion of text or speech from one language into another while preserving the original meaning as accurately as possible.
- For example, translating English text into Malayalam, Hindi, French, or German.

Example

- SOURCE LANGUAGE — ENGLISH

  Good morning. How are you?
- TARGET LANGUAGE — FRENCH

  Bonjour. Comment allez-vous ?
- The translation conveys the meaning of the original sentence in another language.

TYPES OF MACHINE TRANSLATION

- **Rule-based machine translation:** Uses manually defined grammatical rules and dictionaries.
- **Statistical machine translation:** Uses statistical patterns learned from parallel text.
- **Neural machine translation:** Uses neural networks to learn mappings between languages.
- **Transformer-based translation:** Uses attention-based architectures to model contextual relationships and generate translations.

Real-world applications

- Google Translate and Microsoft Translator.
- Translation of websites and documents.
- Multilingual customer support.
- Translation of educational resources.
- Cross-language communication.
- Translation of government and public information.

Limitation: Translation quality may decrease for low-resource languages, idioms, cultural expressions, ambiguous sentences, and specialized technical terminology.